

- AFMCP learners, how are you doing out there? This far into the program, If you're like I was, your brain is probably getting pretty full. So this is actually gonna be a more fun session, or at least I think it's a lot more fun, because we're gonna actually activate some of the knowledge that you're gaining and show you how it applies to your clinical practice. With this session, I have no specific disclosures. These are our learning objectives for the session, and these are the clinical skills and performance objectives. Take a look at your leisure. What we're gonna focus on today is modifiable lifestyle factors and how they can affect the various nodes on the matrix. Here is that functional medicine matrix. So remember, along the side of the matrix are our antecedents, triggers, and mediators. Very important to get a contextual sense of what's going on with our patients. In the center, and the most important here in the matrix, are the nodes: assimilation, defense and repair, energy, mitigation and clearance, transport, communication, and structural integrity. And of course, let's not forget the mental, emotional, and spiritual aspects of taking care of our clients and patients. Today we're gonna focus on the bottom part of this matrix which are the modifiable lifestyle factors: sleep, physical activity, nutrition, stressors, and relationships. And we are gonna talk in this session about how each of these modifiable lifestyle factors affects the specific nodes that we're dealing with. And we're gonna start with assimilation. So when we look at assimilation and sleep, intuitively, we might not think that sleep affects how our body takes things on, but it has a very central role. Sleep can actually be a mediator of dysbiosis. And by that I mean poor sleep can actually affect dysbiosis. Suboptimal sleep may indirectly affect maldigestion and malabsorption by disrupting our microbiome, which as you know, at this point in time, is hugely important for assimilation. Sleep disturbances are also associated with IBS, GERD, and several other GI disturbances. And in an observational study, people with shortened sleep actually had about a 32% increased odds of constipation; and GI transport time, as we know, is so important. Physical activity can actually mediate dysbiosis as well. So here we have on the right side of the screen someone who's doing a lot of exercise. And high-intensity strenuous exercise can actually increase our stress hormones, which affect our nervous system activity, resulting in changes in gut motility, changes in gut transit time, and nutrient transport activity; which, on the other hand, poor activity or lack of physical activity can slow down transit time and increase our risk of dysbiosis and maldigestion. Stress: who doesn't have stress in their lives? Well, stress definitely can affect dysbiosis. So when we're experiencing a lot of stress, what does that do to our GI system? What does that do to our body? Well, it can alter our GI motility, impair nutrient absorption. It leads to sympathetic nervous system activation. It gets us out of that rest-repair-digest mode of the parasympathetics into that fight or flight mode, which makes it very difficult for our body to assimilate. The enteric nervous system gets dysfunctional as well. It affects motility, it affects the way that our brain links to our gut to assimilate the

nutrition that we're taking in. And stress can actually decrease our stomach acid leading to maldigestion issues as well. Now, on the opposite side, meditation has wonderful effects. It puts us back in that parasympathetic rest-repair-digest mode that our bodies really need to be in for optimal health. So how does meditation help? Well, it improves that cephalic phase of digestion, that anticipatory nervous system that helps us regulate digestion. It supports downstream digestive processes by activating that parasympathetic nervous system, getting our body ready and calm to be able to assimilate what we're taking in. Stress can also affect the gut microbiome. Observational research suggests that psychosocial stress is a risk factor for psychiatric illness, and we see a lot of that in our medical training. But did you know that psychosocial stress can also have a similar role in dysbiosis? Early-life stress alters the enteric microbiome with effects persisting into adulthood. So kids who have a lot of stress have persistent microbiome effects into adulthood. And there was even a study done that looked at people who had a significant stress response and major depressive disorder, and they were found to have a lower abundance of these bacteria. So stress definitely links to the microbiome, which of course is a central core concept as we talk about assimilation. And relationships, that last of the modifiable risk factors, definitely can mediate dysbiosis. It's been found that spouses, partners, will have more similar microbiota and more bacteria in common than siblings, even when accounting for dietary factors. Married individuals will actually harbor microbiomal communities of greater diversity and richness relative to those living alone. So if you live in a community, the community of bacteria in your gut is actually stronger. Couples who report closer relationships actually have the greatest microbiome diversity. And on the flip side of that, if you are in a dysfunctional partner relationship, that can cause you to have an altered microbiome as well. And there's a whole new area of research looking into that. And how does this happen that these relationships mediate better microbiomal diversity? Well, it could be cohabitation, physical contact, or shared lifestyle habits. But we're still learning a lot about this as the research continues. So again, as we step back a bit and look at that assimilation node, and specifically the modifiable lifestyle factors, poor sleep, inactivity, chronic stress, not being in a great relationship can lead to a lot of imbalances in our body that are manifested by these medical conditions like gastroesophageal reflux disease, motility problems, irritable bowel, and dysbiosis. Now let's move on to defense and repair. So as we look at sleep disturbances impacting defense and repair, there have been studies that show that acute sleep deprivation stimulates inflammatory gene expression and pro-inflammatory cytokines. Think about how you think after you don't sleep well for an night or two, right? You probably feel a little bit foggy while there may be pro-inflammatory cytokines around in your system that are causing a little bit of cognitive dysfunction. Evidence suggests that prolonged sleep disturbance can lead to an increase in inflammatory markers like CRP and IL-6. And by the way, keep CRP and IL-6 close by

in your mind, because we are gonna talk about them a lot. Healthy volunteers, speaking of sleep deprivation, were randomly assigned a study to limit sleep for four hours per night for 12 days, and they showed significantly elevated IL-6 compared to controls who were allowed to sleep eight hours per night. So really short sleep disturbances can have these pro-inflammatory effects; probably not something that most conventional docs or healthcare people are thinking about when we approach our patients. Physical activity of course modulates defense and repair. There are several studies now that show that resistance training and strength training both show a decrease in these pro-inflammatory cytokines like IL-6 and TNF alpha, but you don't have to lift heavy to get the benefit of that. Even moderate intensity exercise can show the same IL-6 and possibly CRP improvements in this particular study looking at previously sedentary old men. So the key is do some movement, and we should be able to see an improvement in these inflammatory markers. Sugar consumption, as we talk about modifiable lifestyle factors, nutrition is key. That's why it's the center of the five modifiable risk factors. But sugar consumption definitely affects these pro-inflammatory mediators. High intake of dietary sugars actually increase inflammatory cytokine production in the adipose tissue and liver; high intake of these dietary sugars stimulate our immune response to be more avid, to be more aware, and to attack things that it normally wouldn't attack; we get Th17 differentiation via mitochondrial reactive oxygen species; we have a higher increase in toll-like receptor 4 in macrophages, and that is the signaling molecule that's scouring, looking for things to attack, right? We call these PAMPs and DAMPs as you'll learn in the subsequent lectures. And finally, increased dietary sugar leads to a higher degree of gut permeability. And leaky gut, as you are learning, is so central. It's a core principle to a lot of the pathology that we talk about here at the IFM course. And here's something that I didn't realize until recently, but loneliness can actually modulate defense and repair as well. There's a difference between being alone and loneliness, but it's that emotional response to being lonely or to being by yourself that really puts people in a negative spin. Loneliness is correlated with increased inflammatory cytokines. People who self-report greater degrees of loneliness have higher levels of IL-6. In a study that was done, people who reported less social connection actually had a heightened TNF alpha response when exposed to bacterial endotoxin. So we're not just talking about mental response, but actual physiologic response to an endotoxin was increased by that sensation of feeling lonely. And finally, loneliness has also been associated with decreased antiviral immunity, increased antibody titers to certain viruses like Epstein-Barr, CMV, and HPV, and decreased immunoglobulin levels like IGA, IGG, and IGM. So again, as we get back to the matrix, looking at our defense and repair, we see a significant response in inflammatory cytokines and our body's ability to mount regulated immune response when we see these dysfunctions in the lifestyle factors: poor sleep, high sugar diet, loneliness, all of these contribute. Moving along to my next favorite

node of communication, how do these modifiable risk factors affect our communication? Well, remember, when we talk about communication, we talk about how the body speaks to itself: autocrine, endocrine, and paracrine responses. And in particular, when we look at sleep in women, the effect on estrogen of sleep deprivation, this was a pretty sentinel study looking at just under 300 women, mean estradiol concentrations increased by 3.9% for every hour increase in daily sleep. And this was measured by sleep diaries. Interestingly, as well, mean luteal phase progesterone concentrations increased by 9.4% for every hour of increased sleep. So these are pretty significant bumps in estrogen, or estradiol and progesterone levels, just by something as simple as sleep, right? And in women who reported night work or shift work, they were found to have lower testosterone levels compared to employed women who worked during the daytime or did not work night shifts. So again, we have to take into account, when you're speaking to your patients, looking at antecedents, triggers, and mediators, understand their life, understand the context of their health, understand what they're trying to accomplish and how they have to go about their daily life, and make sure that's a part of your mental process of putting together this patient's health picture. Remember, these hormonal changes associated with inadequate sleep have the potential to significantly impact GYN health. But it's not just women; men also see disruptions in their hormones as well. In this randomized controlled trial of gentlemen with obstructive sleep apnea versus healthy volunteers, researchers found that there was an association between sleep apnea/sleep disordered breathing, and lower levels of both total and free testosterone. As we talk about fertility issues, there was also an association with sleep apnea and lower levels of healthy sperm and lower sperm concentration. Finally, lower levels of testosterone were associated with episodes of sleep deprivation as well. Fortunately, partial sleep deprivation didn't lead to these changes, but we have to talk about sleep and we have to optimize sleep to make sure that our patients are getting the optimal health that they can. Physical activity as well is so important to that communication node, as we see in this study looking at cortisol, which is the stress hormone that we'll learn so much about through the course. In this small study of 20 participants at about 40 years of age, they underwent this study where, I keep saying the word study, sorry, and physical activity is involved in the communication node as well. For those of you who are doing the course right now, cortisol comes up frequently when we talk about the HPA axis; well, in this study of 20 people, we looked at the experience of work and an intervention of treadmill desk, standing desk, or standing-only on cortisol. And surprisingly, what they found was that people who participated in the study, who were either at a standing desk or a treadmill desk, had significantly lower afternoon salivary cortisol levels than people who were in the sitting-only group. So that activity, that movement, the physical activity of even just standing led to better communication or healthier communication within that patient's system. Thyroid hormone as well can respond to stressful

figures. Thyroid hormone as well can respond to stressful triggers. In this study, among critical care workers and match controls, people who were exposed to greater work-related stress had higher IL-6, we've already talked about IL-6, but also higher levels of TSH and LH, and significantly lower levels of free T4 when compared to their control group. So the stress at work affected their thyroid function as well. And when we look at outside of healthcare workers, looking at non-healthcare folks, chronic HPA axis activation has been shown to have deleterious effects as well. So there is this study that looked at people who had been affected by an earthquake, and they were found to have an increased incidence of high cortisol post-earthquake. And, as we know, these stressful events can trigger the HPT axis, and actually had higher levels of altered TSH. So thyroid hormone is one of the most sensitive hormones and is definitely affected by stress. And as we talk about communication, stress and relationships can definitely mediate that communication node. And in particular, I think we have to be cognizant of adverse childhood events, or ACEs. In this study, women who had four or more of these adverse childhood effects were almost two and a half times more likely to experience premenstrual dysphoric disorder and premenstrual syndromes compared to women with no ACEs. Lower age of menarche was associated with a higher odds of experiencing fear, distress, and externalizing disorders, and that correlated to their ACE scores as well. When we look at miscarriages, women with three or more ACEs are two times as likely to have a single miscarriage, and over three times more likely to experience recurrent miscarriages compared to women who have not experienced a miscarriage. So as we do our intake, again, it is not always a comfortable question, but it's an important question to ask for our patients. And then finally, among women aged 18 to 45, a study included that each ACE increase the risk of fertility difficulties by about 9%, and a reduced fecundability by 32%. So please, please, please ask these questions of your patients, because they need to let you know so you can take ideal care of them. So as we come back to the communication node, looking at these modifiable lifestyle factors, poor sleep, occupational stressors, higher ACE scores can have significant effects on the communication node; irregular menstrual cycles, fertility, both for men and women, and then thyroid dysfunction can be a result of these communication issues. And let's move on to transport. As a cardiologist, this is probably my favorite node. I say that each one is my favorite, but transport really is my favorite, right? So let's take a look at this. So when we look at the modifiable lifestyle factor of sleep, sleep has a significant effect on transport dysfunctions. Blood pressure, it was found that people who had inadequate sleep often had higher blood pressure than those who reported good sleep. Sleep apnea, as we touched on briefly before, has been significantly associated with sleepiness, inflammation, and insulin resistance, and showed biochemical markers like high-sensitivity CRP, IL-6, leptin, and a higher degree of insulin resistance than controls. And as we talk about lipids, when we looked at a meta-analysis, patients with obstructive sleep apnea actually had

increased propensity towards dyslipidemia, specifically high total cholesterol, high LDL, high triglycerides, and low HDL. And if those sound familiar, a lot of those are factors in the metabolic syndrome. So as we look at physical activity, everybody knows that physical activity is good for your heart, right? But why? Well, these are some of the biochemical markers and reasons why, blood pressure. So in a study where we put patients through aerobic exercise, we saw an 8% reduction of systolic blood pressure and almost 5% or a five-point reduction in diastolic blood pressure, and a significantly lower heart rate. And as we look at the evidence for blood pressure medicines, this is on par, this change in blood pressure is on par for what we expect of a single blood pressure agent. So physical activity can improve blood pressure as much as a blood pressure medication can. And we'll talk more about that as the course continues. Getting back to lipids, exercise can improve lipids, particularly can lower triglycerides and can lead to a modest increase in HDL and Apolipoprotein A-I. And then finally, when we talk about metabolic dysfunction, number one, please do the cardiometabolic module; it is my favorite. But getting back to this, as we look at metabolic syndrome and type 2 diabetes, aerobic activity becomes very important, as does resistance activity. So in a supervised aerobic and resistance exercise course, patients saw a significant lower hemoglobin A1c than compared to the no exercise group, almost 0.3% lower. And a combined aerobic plus resistance exercise program produced even better hemoglobin A1c control than just aerobic training alone. So you have to add resistance training to the exercise prescription that we're giving our patients, and we'll talk more about that in other modules. Perceived stress and the transport nodes: so we're not talking about actual stress but how much stress do you feel you have. This was a very instructive study that looked at almost 2,000 Black participants without hypertension at baseline. And during that time, about 48.5% were found to have hypertension, but the perceived stress that they had actually affected their blood pressure responses. So in individuals with low perceived stress, their incidence of hypertension was about 30%. and people with moderate perceived stress, about 34.6%. And in individuals who perceived that they had a high amount of stress, the incidence of hypertension was 38.2%. So this is a situation where we weren't looking at how much actual stress they had but how do they perceive it, how did that stress hit their bodies? And you can see the swing between 38.2% of people who felt they had a lot of stress and people who were managing their stress a bit better, 30.6%. So almost a 10% swing in the incidence of blood pressure just by mindset alone. Loneliness has been, again, described as such a significant public health issue. But loneliness has been associated with hypertension as well. Greater degrees of loneliness are predictive of exaggerated blood pressure response and exaggerated inflammatory response to acute stress. In a systematic review, loneliness was associated with a higher baseline blood pressure; lonely individuals were also found to have a lower cardiac output and lower heart rate compared to non-lonely people. So what does that

mean? Well, loneliness may cause a blunted response to stress compared to non-lonely people. And getting back to the study that we just referenced, lonelier individuals may perceive certain stressors as more stressful, so they will have a higher degree of perceived stress. So we want to make sure that when you're talking to your patients, we want to address that feeling of community and if there are any feelings of loneliness for them. Connecting back to our transport node, poor sleep, lack of exercise, multiple stressors, and loneliness have all been associated with hypertension. Thank you for listening so far. My colleague, Dr. Regina Druz, will now take you through the rest of the nodes in the functional medicine matrix.